

# Error Calculation & Measurement Uncertainty

*Method Dossier | Basic Level*

## 1. Method Description

Systematic errors (bias) vs random errors (noise). Gaussian error propagation:  $\sigma_Y = \sqrt{\sum (df/dX_i)^2 \cdot \sigma_{X_i}^2}$ .  
For addition: add squares. For multiplication: add relative squares.

## 2. Examples

Physics:  $g = 2h/t^2$ ,  $\sigma_g/g = \sqrt{0.005^2 + (2 \cdot 0.01)^2} = 2.1\%$

Medicine: Blood pressure 138/142/134 mmHg. Mean=138, SD=4.0, 95% CI = 138+/-9.9 mmHg (n=3 is insufficient).

Sports: Jump force 5 trials, SD=120N, relative error = 2.2% -> improve to 1.6% with 10 trials.

## 3. Assumptions

- Random errors normally distributed
- Input errors independent (or include covariance)
- Systematic errors corrected or reported separately

## 4. Reporting

Value +/- uncertainty (unit) at k=2 (95% CI). Relative uncertainty in %.

## 5. Code Examples

Python:

```
import numpy as np; from scipy.stats import t
m,s = np.mean(data), np.std(data, ddof=1)
ci = t.interval(0.95, df=len(data)-1, loc=m, scale=s/np.sqrt(len(data)))
```

## 6. Extension

GUM standard: Type A (statistical) + Type B (other) uncertainty. Monte Carlo for nonlinear functions.

## 7. Exercises

1. Reaction times: 220,245,235,210,240 ms. Mean, SD, SE, 95% CI (t=2.776, df=4). How many trials for SE<5ms?

2. Density  $\rho = m/V$  from  $m=100\pm 1g$ ,  $V=50\pm 2cm^3$ .  $\rho$  with error propagation. Relative error?